

Applying KVL on the the circuit in Fig 1.:

$$V_{\text{dc}} = r_L i_L + L \dot{i}_L + v_C \Rightarrow \boxed{\dot{i}_L = \frac{1}{L} (-r_L i_L - v_C + V_{\text{dc}})} \quad (1)$$

The capacitor current-voltage relation is:

$$i_L = C \dot{v}_C \Rightarrow \boxed{\dot{v}_C = \frac{1}{C} i_L} \quad (2)$$

Eliminate v_C in (1) by differentiating wrt time and using (2) to get a 2nd-order ODE in i_L :

$$\ddot{i}_L = \frac{1}{L} (-r_L \dot{i}_L - \dot{v}_C) = -\frac{r_L}{L} \dot{i}_L - \frac{1}{LC} i_L$$

$$\boxed{\ddot{i}_L + \frac{r_L}{L} \dot{i}_L + \frac{1}{LC} i_L = 0}$$

Characteristic polynomial and comparison with the canonical 2nd-order form:

$$s^2 + \frac{r_L}{L} s + \frac{1}{LC} = 0 \quad \Longleftrightarrow \quad s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\boxed{\omega_n = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{r_L}{2} \sqrt{\frac{C}{L}}}$$